

- 1 (a)** Fig. 1.1 is a drawing of a longitudinal section (LS) of a human kidney.

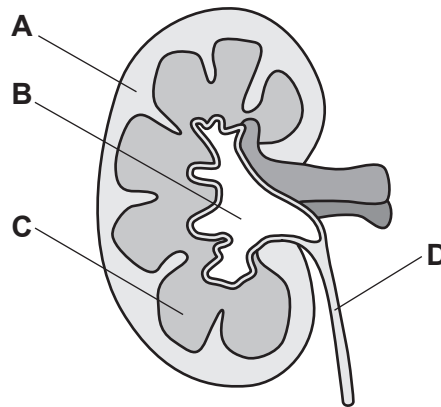


Fig. 1.1

Use the letters **A**, **B**, **C** and **D** in Fig. 1.1 to complete Table 1.1.

Each letter may be used once, more than once or not at all.

For each description, list **all** the letters that are correct.

Table 1.1

description	region of kidney
location of loops of Henle	
location of Bowman's capsules	
location of glomeruli	
contains urine at final concentration	

[4]

- (b)** The volume and water potential of the urine produced by the kidney vary according to the water potential of the blood. This is a result of osmoregulation.

Describe the role of aquaporins in osmoregulation.

..... [3]

[3]

- (c) Describe the role of the brain in osmoregulation when the water potential of the blood **increases** above the set point.

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..... [3]

[Total: 10]

Question 2 starts on page 5.